



Green University of Bangladesh

Department of Computer Science and Engineering (CSE)

Hotel Booking Management System

Course Title: Object Oriented Programming Lab

Course Code: CSE-202

Section: D2

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Submission Date: 15/05/2024

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<u>Lab Project Proposal Status</u>	
Marks:	Signature:
Comments:	Date:

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0.1 TITLE OF THE PROJECT PROPOSAL

Hotel Booking Management System

0.2 INTRODUCTION

The goal of the Hotel Booking Management System project is to completely transform the administration and facilitation of hotel reservations. Our project uses cutting-edge database technology to build a solid system that guarantees effective management of bookings, reservations, and customer data. We work to improve the hotel industry's user experience, operational efficiency, and resource usage through careful design and implementation.

0.3 OBJECTIVES/AIMS

Creating a Java Swing-based application that enables users to effectively manage hotel reservations is the goal of the Hotel Booking Management System project. Customers and administrators alike will find this system's user-friendly interface convenient for carrying out several hotel reservation-related duties.

Key Objectives:

- **User-friendly Interface:** Design a visually appealing and intuitive graphical interface using Java Swing components to facilitate easy interaction for users.
- **Booking Management:** Enable customers to view available rooms, make reservations, and manage their bookings. Allow administrators to add, modify, and cancel bookings as needed.
- **Database Integration:** Implement JDBC to connect the application to a backend database (e.g., MySQL) for storing and retrieving booking information securely.
- **Authentication and Authorization:** Implement user authentication mechanisms to ensure secure access to the system. Differentiate between customers and administrators with appropriate privileges.

- **Error Handling and Validation:** Implement robust error handling and input validation to ensure data integrity and prevent unauthorized access or misuse of the system.

By achieving these objectives, the Hotel Booking Management System will streamline the process of managing hotel reservations, enhance user experience, and contribute to the efficient operation of hotel booking services.

0.4 PROPOSED FEATURES

- **Booking Management:** Enable users to create, modify, and cancel bookings. Include features to view booking history, manage guest details, and handle special requests.
- **Billing and Payment:** Integrate payment processing functionality to handle booking payments securely. Generate invoices and receipts for completed bookings.
- **Guest Feedback:** Allow guests to provide feedback and ratings after their stay. Use this data to improve service quality and guest satisfaction.
- **Responsive Design:** Design the user interface using Java Swing components that are responsive and intuitive, providing a seamless user experience across different screen sizes.

0.5 METHODOLOGY

- **Requirements Gathering:** Begin by gathering and analyzing the requirements for the hotel booking system. Identify the essential features such as booking rooms, managing guest information, handling reservations, and generating reports.
- **System Design:** Design the architecture of the application. Define the classes and modules needed to implement the system. Create a database schema to store information related to hotels, rooms, guests, bookings, etc.

- **Applying Business Logic:** putting the hotel booking system's business logic into practice by writing Java code. This covers features including creating new reservations, managing reservations, changing guest information, and billing.

0.6 TOOLS AND TECHNOLOGIES

- **Operating System:** Windows 11 Home.
- **Integrated Development Environment:** Netbeans-20
- **Database Managment System :** MySQL
- **Database Connector :** JDBC Driver
- **Documentation:** LaTeX.

0.7 CONCLUSION

In conclusion, utilizing Java with Java Swing to construct a hotel booking management system provides a strong and adaptable way to make a desktop application that is easy to use. Java Swing makes it easier to create a sophisticated graphical user interface (GUI) with a variety of components, such as buttons, text fields, and tables, while Java itself offers the resilience and platform freedom required for such projects. Data integrity and scalability are ensured by the system's seamless interaction with a backend database through the use of JDBC (Java Database Connectivity) to store and retrieve booking information. The project highlights Java's adaptability in creating sophisticated yet user-friendly apps appropriate for a range of settings.